Project – 1

Title: Railway_Guages.

Question:

Here are 5 well-structured, end-to-end scenarios (with preprocessing → analysis → visualization), balanced by difficulty as you requested:

Scenario 1: Basic Data Loading & Cleaning.

First, we have to import this CSV file into Pandas library. Pandas is a powerful data analysis library providing flexible way to do data manipulation. There are many mathematical and scientific computational features can be leveraged using this library. Let us see some of the features in our exercise.

In order to load the CSV file using Pandas, we need to do the following step.

```
d=pd.read_csv("railway_gauges.csv")
```

Pandas returns as Data frame object. Using data frame object, we can see the top 5 rows using **head()** function and **tail()** function to display last 5 rows in a CSV file. In our above example, we have used **df.head()**, it displays first 5 rows in the loaded CSV. Here I am displaying first 5 & last 5 rows and columns.

	A	B	C	D	E
1	Year	Broad Gau	Metre Gau	Narrow G	Total
2	1964-65	3521	2891	464	6876
3	1965-66	3603	2913	470	6986
4	1966-67	3615	2911	481	7007
5	1967-68	3601	2920	506	7027
6	1968-69	3603	2921	508	7032

46	2008-09	6185	642	203	7030
47	2009-10	6320	543	220	7083
48	2010-11	6419	494	220	7133
49	2011-12	6494	432	220	7146
50	2012-13	6577	379	216	7172

```
# Display first 5 rows and column names
print(d.head())
print(d.columns)
```

Now I am replacing missing values with 0 and converting all gauge columns like Broad Gauge, Metre Gauge, and Narrow Gauge and Total Gauge. to numeric types.

```
# Replace missing values with 0
d = d.replace(np.nan, 0)

# Convert columns to numeric
cols = ['Broad Gauge', 'Metre Gauge', 'Narrow Gauge', 'Total']
for col in cols:
    d[col] = pd.to_numeric(d[col], errors='coerce')
```

Scenario 2: Simple Visualization

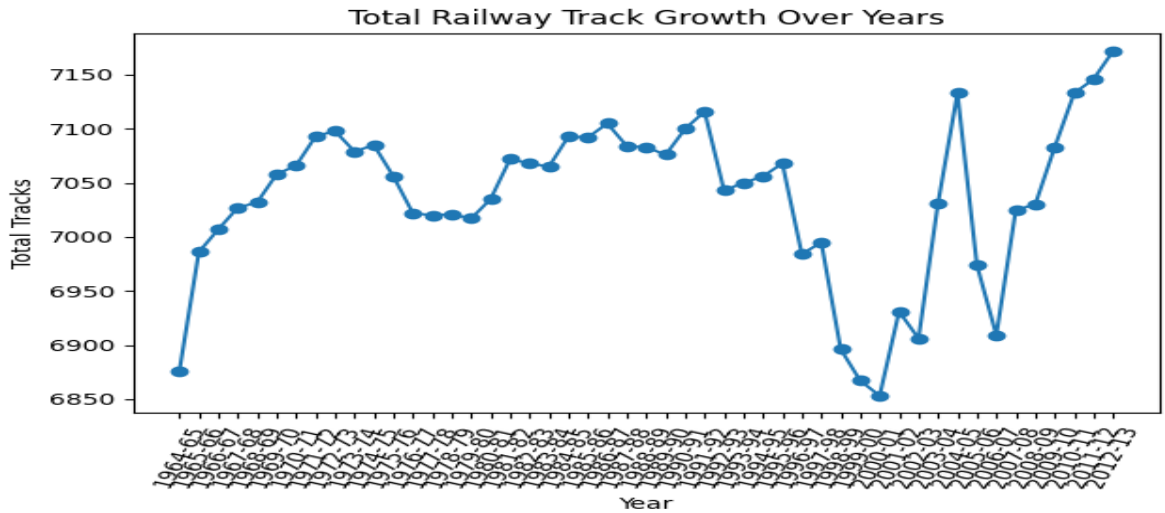
Here, I am Extracting **year** and **total** columns from the Railway Dataset and plotting line graph. That Line graph shows "Total tracks over the years". And the graphs contains **Title** and **X & Y labels** in the graphs.

```
# SCENARIO 2: Line Graph (Total Track Growth)

# Extract Year and Total columns and plot
plt.plot(d['Year'], d['Total'], marker='o')

# Add title and labels
plt.title("Total Railway Track Growth Over Years")
plt.xlabel("Year")
plt.ylabel("Total Tracks")

plt.xticks(rotation=70)
plt.tight_layout()
```



Scenario 3: Filtering + Bar Chart

Here, I am filtering data after 2000 from Railway dataset.

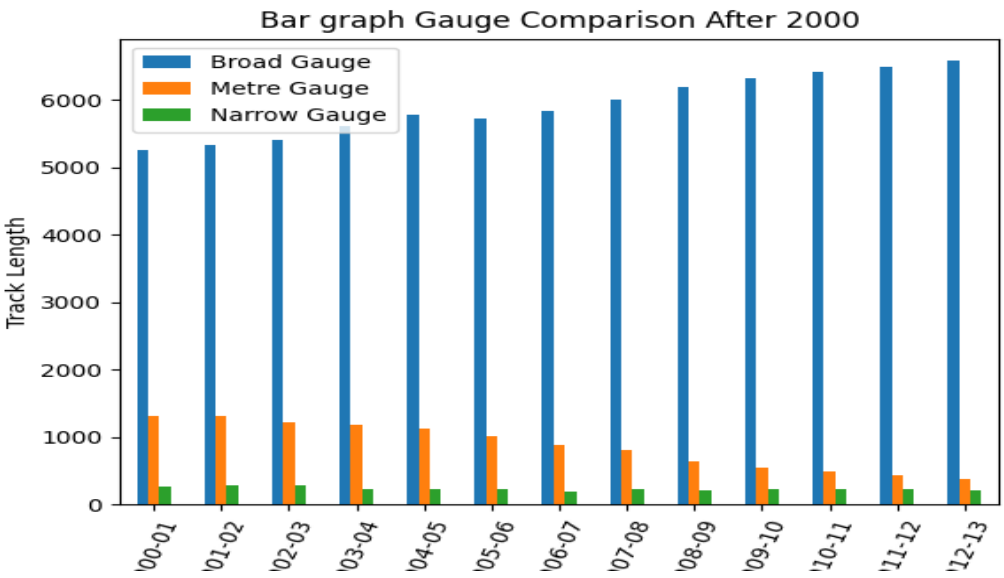
```
# SCENARIO 3: Filtering + Bar Chart

# Filter data for years after 2000
d_modern = d[d['Year'] >= '2000-01']
```

Now I am plotting bar chart comparing Broad Gauge, Metre Gauge, and Narrow Gauge. From dataset. And adding legend to the chart. A **legend** is the small box that tells you what each colour, line, or bar represents.

```
# Plot grouped bar chart
ax = d_modern.plot(x="Year", y=['Broad Gauge', 'Metre Gauge', 'Narrow Gauge']
plt.xticks(rotation=70)
plt.xlabel("Year")
plt.ylabel("Track Length")
plt.title("Bar graph Gauge Comparison After 2000")
plt.legend()

# Save graph
plt.savefig("graphs/gauge_bar.png")
plt.show()
```



<u>Scenario 4</u>: Feature Engineering + Pie Chart

Here, I am Calculate total sum of each gauge across all years using **sum( ) .** And plotting Pie Chart showing percentage contribution and showing percentage labels using **autopct.** Using shadow and explode. These are **styling options** used in **plt.pie()** to make your chart clearer and more attractive .

autopct - autopct is used to automatically display percentage values on each slice.

Shadow - Adds a **shadow effect** behind the pie chart.

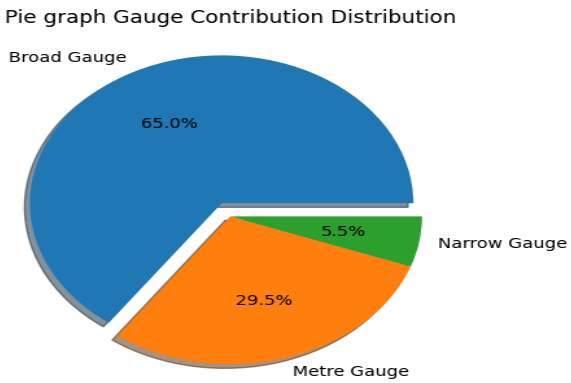
Explode - Separates (pulls out) one or more slices from the pie chart.

```
# SCENARIO 4: Pie Chart (Gauge Contribution)

# Calculate total sum of each gauge
totals = d[['Broad Gauge', 'Metre Gauge', 'Narrow Gauge']].sum()

# Plot pie chart
explode = (0.1, 0, 0)
plt.pie(totals, labels=totals.index, explode = explode, shadow = True, autopct='%1.1f%%')
plt.title("Pie graph Gauge Contribution Distribution")

# Save graph
plt.savefig("graphs/gauge_pie.png")
plt.show()
```



Scenario 5: Advanced Analysis + Multiple Graphs

Here, I am creating new percentage columns for Broad Gauge, Metre Gauge, and Narrow Gauge. And using **numpy** as **np.diff( ).**

np.diff( ) - to computes the difference between consecutive values.

iloc [ ] - is primarily an integer based location search — iloc[0]

```
# SCENARIO 5: Advanced Analysis

# Create percentage columns
d['% Broad'] = (d['Broad Gauge'] / d['Total']) * 100
d['% Metre'] = (d['Metre Gauge'] / d['Total']) * 100
d['% Narrow'] = (d['Narrow Gauge'] / d['Total']) * 100

# Calculate yearly growth using NumPy
d['Growth'] = np.diff(d['Total'], prepend=d['Total'].iloc[0])
```

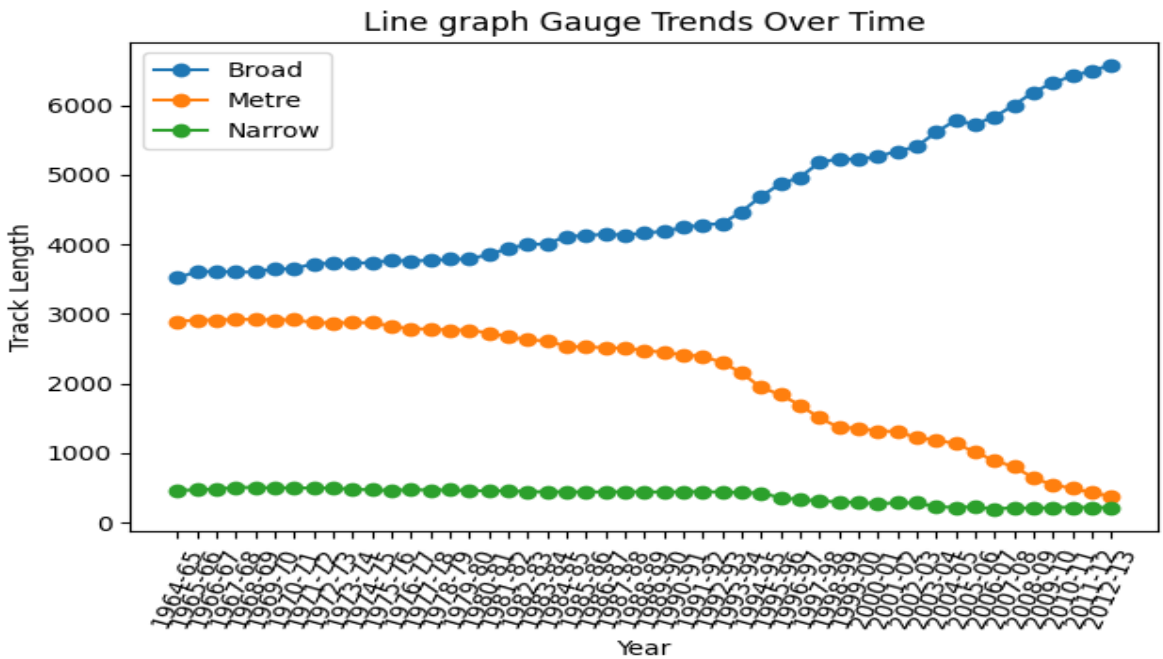
- And plotting Line Graph for all Gauges.

```
# Line graph for all gauges
plt.plot(d['Year'], d['Broad Gauge'], label='Broad', marker = 'o')
plt.plot(d['Year'], d['Metre Gauge'], label='Metre', marker = 'o')
plt.plot(d['Year'], d['Narrow Gauge'], label='Narrow', marker = 'o')

plt.legend()
plt.xticks(rotation=70)
plt.xlabel("Year")
plt.ylabel("Track Length")
plt.title("Line graph Gauge Trends Over Time")

plt.tight_layout()
plt.savefig("graphs/gauge_trends.png")
plt.show()
```

And the output graph look likes in below.



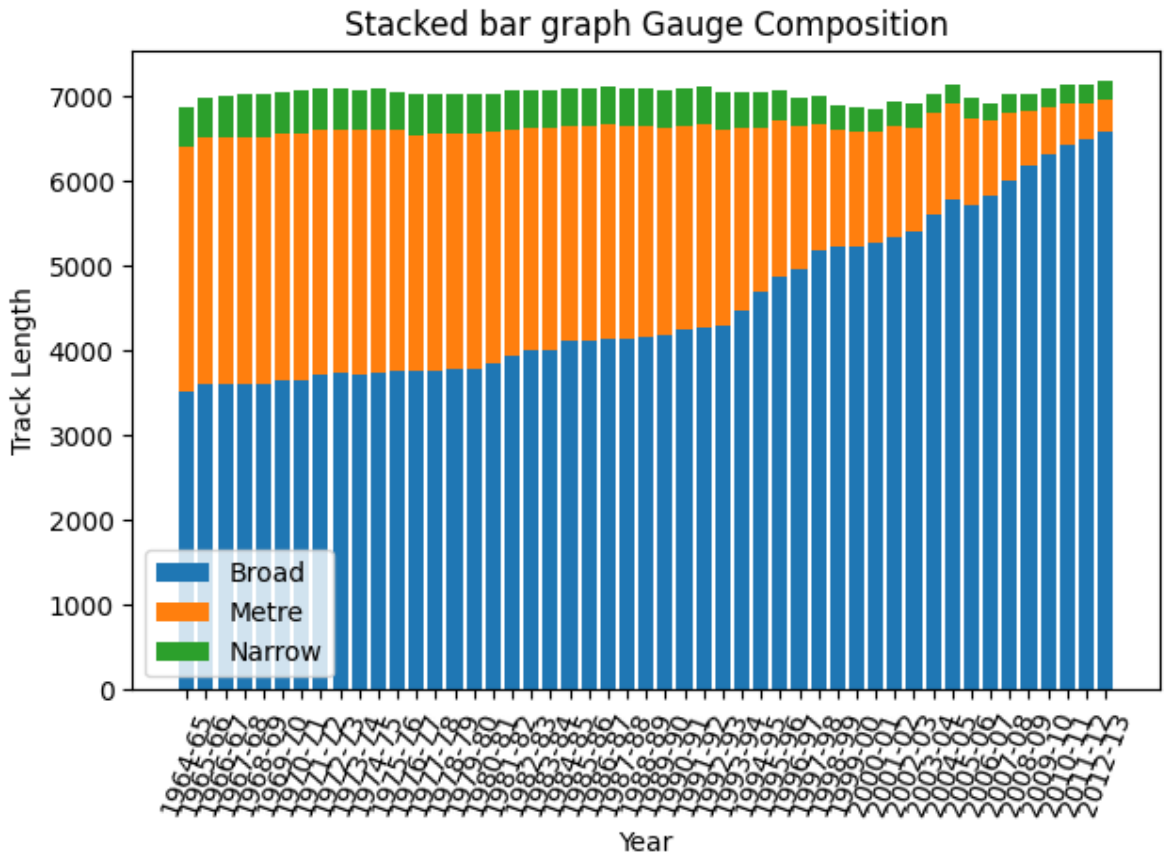
- Stacked Bar chart for showing composition.

```
# Stacked bar chart
plt.bar(d['Year'], d['Broad Gauge'], label='Broad')
plt.bar(d['Year'], d['Metre Gauge'], bottom=d['Broad Gauge'], label='Metre')
plt.bar(d['Year'], d['Narrow Gauge'], bottom=d['Broad Gauge'] + d['Metre Gauge'], label='Narrow')

plt.xticks(rotation=70)
plt.xlabel("Year")
plt.ylabel("Track Length")
plt.title("Stacked bar graph Gauge Composition")
plt.legend()

plt.tight_layout()
plt.savefig("graphs/stacked_chart.png")
plt.show()
```

- And the output graph look likes in below.




```
# Identify highest growth
print("Year with highest growth:")
print(d.loc[d['Growth'].idxmax()])
```

Yes, the railway system is clearly **shifting towards a single dominant gauge**, which is the **Broad Gauge**.

Over the years, the data shows a **steady and continuous increase in Broad Gauge track length**, while both **Metre Gauge and Narrow Gauge are consistently declining**. This trend indicates a systematic effort to convert and standardize railway tracks.

The reduction in Metre and Narrow Gauge is not random—it reflects a **planned modernization strategy**. Broad Gauge offers several advantages such as:

- Higher speed and stability
- Greater load-carrying capacity
- Better compatibility across regions

Because of these benefits, railway authorities have been **gradually replacing older gauges** with Broad Gauge to improve efficiency and connectivity.